

Get the application running locally in a few simple steps.

DQE (Data Quality Engine) pulls live data directly from Opera Cloud / OHIP and compares it against Duetto's own reports — replacing manual Postman calls and spreadsheet diffing with one page. It's built for the DQE team (root-causing Duetto-vs-PMS discrepancies) and the Deployment/Migrations team (pre-go-live configuration checks).

Getting Started

1 Access GitHub

The app lives in a private GitHub repository. You'll need to be added as a collaborator before you can clone it — ask Tyler Linton for access if you don't have it yet.

```
github.com/tylerlinton-png/Data-Quality-Tool-
```

2 Clone the repository

Open a terminal and run:

```
git clone https://github.com/tylerlinton-png/Data-Quality-Tool-.git
```

3 Open the project folder

Move into the project directory:

```
cd Data-Quality-Tool-
```

4 Install dependencies

Requires Python 3.9 or later. Install the required packages:

```
pip3 install -r requirements.txt
```

5 Configuration

No .env file or API keys are needed to install and run the app. Your Oracle OHIP credentials are entered directly in the app itself (Step 1 on the main page) and are saved locally on your machine — never in the codebase.

6 Start the application

Run the app, then open it in your browser:

```
python3 app.py  
→ Open http://localhost:5055
```

Quick Troubleshooting

● “command not found: pip3” or “python3”

Install Python 3.9+ from python.org, or via Homebrew (brew install python3) on Mac.

● “Address already in use” on port 5055

Another copy of the app is already running. Find and stop it, or close the old terminal window, then restart.

● Permission denied cloning the repo

You haven't been added as a GitHub collaborator yet — request access before cloning.

● “Missing required field(s)” when validating credentials

Double-check Hostname, Hotel ID, App Key, and Client ID/Secret are all filled in — all four are required to connect to OHIP.

● A pull or comparison comes back empty

Confirm the date range actually has data for that hotel, and that the External System Code (Step 1) matches what's configured for that property in Opera.

Pull Bookings / Blocks / Folio

WHAT IT DOES

Fetches raw reservations, blocks, or folio data straight from Opera Cloud for a date range — no Postman collection required.

HOW TO USE IT

Pick a data type and date range in the Pull card, click Fetch Data, then search, filter, or download the results as CSV/TSV.

WHY IT MATTERS

Removes the need for manual Postman calls for everyday data checks.

Upload DVA & Compare

WHAT IT DOES

The core analysis engine — compares Duetto's validation export, bookings, and folio data against Opera to score accuracy and root-cause every discrepancy.

HOW TO USE IT

Drop in whichever files you have (all are optional), click Run Analysis, then review the Summary, Discrepancies, and Recommendations tabs.

WHY IT MATTERS

Turns a multi-hour manual spreadsheet comparison into a one-click report with an actual root cause, not just a mismatch.

Pull Deployment Info (Config Snapshot)

WHAT IT DOES

Runs six pre-go-live checks in one click — Opera Cloud Version, Block Status Codes, Transaction Codes, Room Types, Hotel Info, and Rate Plans.

HOW TO USE IT

Click Run Config Snapshot, review each result card, and use Download All for a single Excel workbook with everything.

WHY IT MATTERS

Gives Deployment/Migrations a fast, complete sanity check before a hotel goes live.

Hotel Stats

WHAT IT DOES

Pulls Revenue Inventory Statistics — rooms sold, revenue, arrivals/departures/cancellations — directly from Opera for a date range.

HOW TO USE IT

Set a date range and click Run Hotel Stats. Results take a little longer since it's a multi-step live job, not an instant call.

WHY IT MATTERS

Answers PMS-side inventory/revenue questions without needing a separate Opera report.

Live API Comparisons

WHAT IT DOES

Compares a live OHIP pull directly against Duetto's own Folio or Bookings report — side by side, including the ones that match.

HOW TO USE IT

Pull live data (or reuse a Browse Data pull), attach the matching Duetto report, then run the analysis — results appear as their own tab.

WHY IT MATTERS

Answers "can I trust what Duetto is showing" without waiting on a fresh export.

Send Feedback

WHAT IT DOES

A direct line to report a bug or suggest an improvement while using the app.

HOW TO USE IT

Click Feedback in the top navigation, describe the issue or idea, and submit.

WHY IT MATTERS

Feedback goes straight to the team building the tool — nothing gets lost in chat.